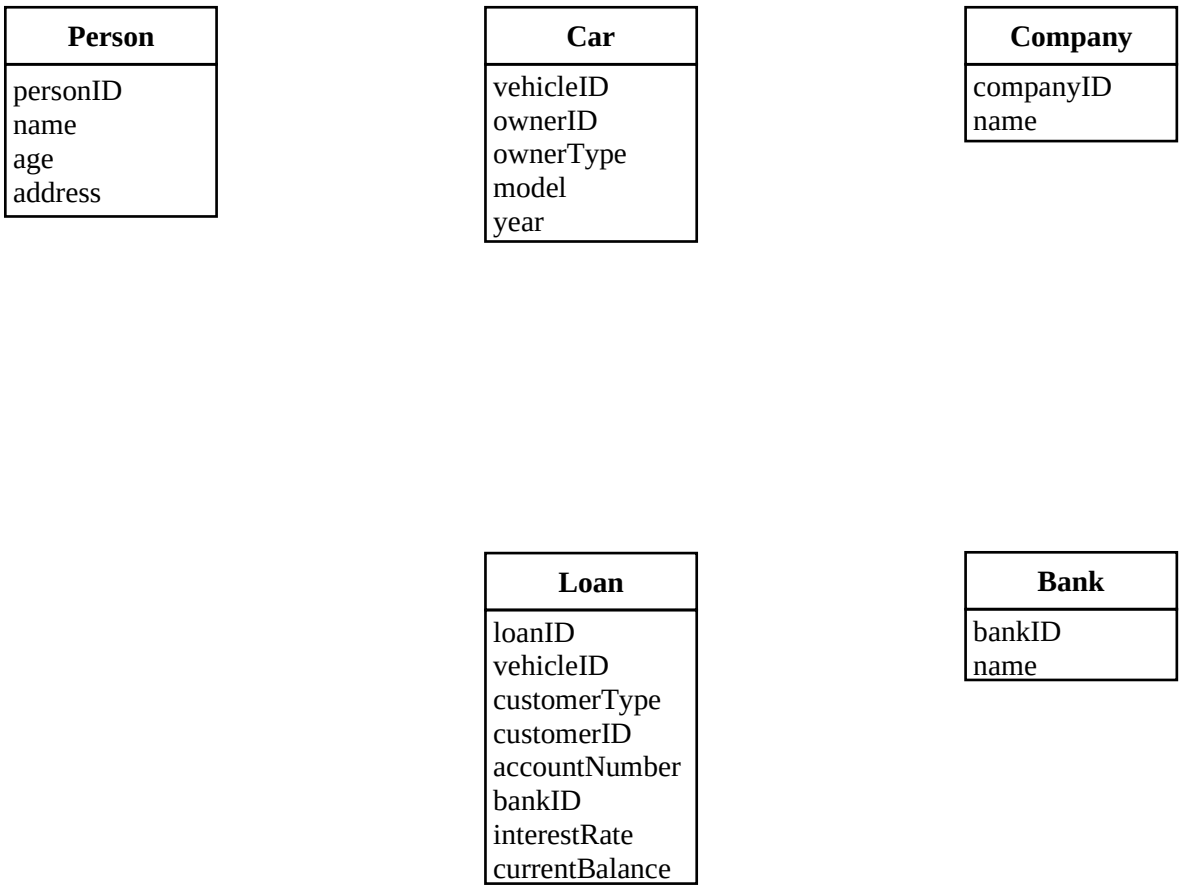
Name: _____ **Date:** _____

Software Engineering Specialization

Exercise: Modeling Software Systems using UML

The classes shown below are required to represent information about car ownership and car loans. However, they have some attributes that are *internal object identifiers (OIDs)*, some of which are being used to represent relationships, and that *should not appear* in a class diagram. All such attributes conveniently have names ending in ID. 🙂

Persons or companies may own cars. The car ownerID is the ID either the person or company that owns the car. A car may have only one owner (person or company). A car may have no loan or multiple loans. A bank provides a loan to a person or a company for the purchase of a car. Only the car owner may obtain a loan on the car. The car owner type and the loan customer type indicate whether the car owner/loan holder is a person or company.

On the class diagram below:

1. **Replace all OIDs with associations.**
2. **Indicate the most likely multiplicities for all associations.**
3. **Cross out any unnecessary attributes.**